

LILY NATASHA WARTMAN

Rice University
Department of Mathematics
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EDUCATION

Rice University

- Ph.D. in Mathematics (Fall 2024 - present).

Scripps College

- B.A. in Mathematics, *with honors*, Minor in Fine Art (2023).
- Senior thesis: *Knot Theory: Exploring the Relationship Between the Alexander Polynomial and Alexander Tribracket Knot Invariants*, Honors Thesis.

PUBLICATIONS AND PREPRINTS

- **To Open or Not to Open: Developing a COVID-19 Model Specific to Small Residential Campuses** (joint with C. Edholm, M. Hohn, N. Falicov, E. Lee). CODEE Journal: Vol. 17, Article 1. (<https://scholarship.claremont.edu/codee/vol17/iss1/1>). January 2024.
- **A Kernelization Algorithm for Minimal Cut Set Enumeration in Logical Graphs** (joint with S. Rubinfeld, L. Taylor). in preparation.
- **Symmetric numerical semigroups and the Kunz cone** (joint with I. Bailly-Hall, A. Brooks, E. Bogle, C. O'Neill, A. Pineau, E. Sullivan, G. Tsoukalas). in preparation.

RESEARCH EXPERIENCE

- **Research Mathematician.** The Johns Hopkins University Applied Physics Laboratory (August 2023 - August 2024).
- **Knot Theory: Alexander Polynomial and Alexander Tribracket.** Senior Thesis, *Scripps College* (Fall 2022 - Spring 2023).
- **Symmetric Numerical Semigroups.** National Science Foundation Research Experiences for Undergraduates (REU) Program, *San Diego State University* (Summer 2022).
- **COVID-19 Epidemiological Mathematical Modeling.** Undergraduate Research Assistant, *Scripps College* (Spring 2021 - Spring 2022).

TALKS

- *Orbifolds and the $84(g-1)$ Theorem.* Current Mathematics Seminar, Rice University, March 2026.
- *Geometric characteristics of numerical semigroups in the Kunz cone.* Joint Mathematics Meeting, AMS, January 2023.
- *Geometric characteristics of numerical semigroups in the Kunz cone.* San Diego State University Department of Mathematics and Statistics Research Talk, San Diego State University, August 2022.
- *Modeling COVID-19 on Campus.* Claremont Math Colloquium Research Talk, The Claremont Colleges, September 2021.

CONFERENCES

- Topology Student Workshop. *Vanderbilt University* (June 2026).
- Rice Dynamics and Geometry Conference. *Rice University* (April 2026).
- Recent Progress in Topological and Geometric Structures in Low Dimensions. *SLMath Institute* (March 2026).
- Introductory Workshop: Topological and Geometric Structures in Low Dimensions & Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups. *SLMath Institute* (January 2026).
- Pathways Workshop: Topological and Geometric Structures in Low Dimensions & Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups. *SLMath Institute* (January 2026).
- Nebraska Conference for Undergraduate Women in Mathematics, *University of Nebraska - Lincoln* (January 2023).
- Joint Mathematics Meeting (January 2023).
- Graduate Research Opportunities for Women (GROW) Conference, *Duke University* (October 2022).

TEACHING EXPERIENCE

Rice University

- Instructor of Record, Multivariable Calculus (Summer 2026)
- Teaching Assistant, Graduate Geometric Topology (Fall 2025), Graduate Complex Analysis (Spring 2026).

Scripps College

- Teaching Assistant, Modeling and Simulation, Calculus I, Methods in Modern Modeling (Fall 2021 - Fall 2022).
- Math Tutor, MathSpot (Fall 2022 - Spring 2023)

University of Washington

- Teaching Assistant, Summer Institute of Mathematics at the University of Washington (SIMUW) (Summer 2021).

LEADERSHIP

- Laspa Fellow, Laspa Center for Leadership, *Scripps College* (September 2022 - May 2023).
- Treasurer, Scripps Associated Students Government, *Scripps College* (May 2021 - May 2022)